

KWESHUN

Academic Competition & Question Authoring Platform

Complete Architecture, Functional Feature Guide & Documentation • v1.0.0

1. Executive Summary & Core Purpose

KWESHUN is a modern gamified academic competition web application designed for students, educators, and researchers. Unlike passive flashcards, Kweshun transforms exam preparation and peer learning into a competitive esports-style arena with real-time peer duels, rapid-fire lightning sprints, and peer-reviewed question authoring.

Scholars earn Distinction Points (DP), maintain daily active flame streaks, unlock tiered honors (from Scholar Tier up to Grandmaster), and climb global and school leaderboards through verified academic merit.

2. Academic Disciplines & Content Repository

The platform comes pre-loaded with curated, peer-reviewed question banks across 6 core academic subjects:

- **Cell Biology & Mitosis:** Cell cycle checkpoints, meiosis crossover, cyclin-CDK complexes, and organelles.
- **Organic Chemistry Reactions:** SN1/SN2 mechanisms, electrophilic aromatic substitution, and reagents.
- **Electromagnetism & Circuits:** Gauss's law, Maxwell equations, RLC circuit dynamics, and Lorentz force.
- **Data Structures & Algorithms:** Balanced search trees, dynamic programming, graph traversal, and asymptotic complexity.
- **Microeconomics & Game Theory:** Nash equilibrium, consumer surplus, price elasticity, and oligopoly.
- **World History 1900–1950:** Treaty of Versailles, League of Nations, interwar economics, and WWII geopolitics.

3. Complete Platform Feature Guide (Part I)

A. Home Dashboard [Route: [/#](#) or [/#/home](#)]

What it does: Personal scholar headquarters displaying live DP balance, global rank, battle win rate (W/L ratio), and daily streak flame.

How to use: Click "Daily Lightning" for rapid sprints, "Quick Match" for duels, or "Practice" on any deck to launch interactive practice.

B. Question Quality Engine [Route: [/#/questions](#)]

What it does: Authoring and peer-review hub featuring a deterministic 5-dimension automated quality calculator (Originality, Factual, Balance, Distinction, Citations).

How to use: Click "Author Question", fill in 4 choices, explanation, and citation. View the live 0-100 quality score and choose Direct Publish (+30 DP) or Submit.

3. Complete Platform Feature Guide (Part II)

C. Peer Battle Arena [Route: [/battles](#)]

What it does: 5-Round Head-to-Head competitive quiz matches with a 15-second countdown timer, live score calculation, and post-duel confetti results.

How to use: Click "Instant Quick Match" or "Challenge Peer". Answer within 15s per round. View win/loss outcomes, DP gains, and rematch options.

D. Decks & Practice Repository [Route: [/decks](#)]

What it does: Academic catalog allowing topic-by-topic mastery tracking and interactive step-by-step quiz practice sessions.

How to use: Filter decks by subject (Science, Math, CS, Economics, History) or search keywords. Click "Practice Deck" to earn +10 DP per correct answer.

E. Friends & Peer Network [Route: [/friends](#)]

What it does: Social connection center showing online status, streaks, win rates, and incoming connection requests.

How to use: Click "Invite Peers" to generate a referral link or email direct invites (+50 DP). Click "Challenge" on any friend card to start a custom duel.

F. Distinction Leaderboard [Route: [/leaderboard](#)]

What it does: Dynamic ranking system with Top-3 scholar podium (#1 Gold, #2 Silver, #3 Bronze) and full ranked table.

How to use: Toggle between "Global" vs "Friends Only" and "Monthly" vs "All Time". Click "Challenge" next to any ranked scholar to initiate a match.

G. Notification Center [Route: [/notifications](#)]

What it does: Real-time actionable alert inbox for challenge invites, peer-review approvals, and DP milestones.

How to use: Click "Action" on any notification to directly accept battle invites or review questions. Click "Mark all as read" to clear unread badges.

H. Scholar Profile & Ledger [Route: [/profile/:id](#)]

What it does: Complete academic curriculum vitae showing badges, win/loss stats, profile editor, and full Distinction Points transaction history.

How to use: Click "Edit Profile" to modify bio, handle, and university. Inspect the Distinction Ledger table to trace every earned DP transaction.

4. Technical Architecture & State Management

A. Frontend Tech Stack

- **Framework:** React 19 with Vite 8 for instant sub-second hot-reloading and lightweight production builds.
- **Routing:** React Router (HashRouter) enabling deep URL linking, browser back/forward history, and static host compatibility.
- **Styling:** Tailwind CSS v4 with custom dark navy tokens (#090d16, #0d121f, #152037), mint (#0df2c9), and purple accents.
- **Icons & FX:** Lucide React icon system and Canvas Confetti celebration effects on battle victories and practice completions.

B. State & Data Layer Architecture

- **DataStore (services/dataStore.js):** Centralized reactive state engine synchronizing Users, Decks, Questions, Battles, Friends, Notifications, and Activities with localStorage persistence.
- **Quality Scorer (services/qualityScorer.js):** Deterministic mathematical scoring algorithm computing weighted composite grades (A+, A, B, C) and actionable recommendations.
- **Context Providers:** ThemeProvider (Dark/Light mode), ToastProvider (reactive alerts), AuthProvider (user session), and GameProvider (game actions).

5. Interactive Verification & Flow Checklist

All 7 key user journeys have been fully implemented, integrated, and verified:

- **Flow 1 (Authentication & Profile):** Sign in with academic credentials -> switch demo profiles -> edit bio and institution -> verify ledger.
- **Flow 2 (Question Creation & Quality):** Open Question Modal -> fill prompt & options -> observe live 5-dimension quality bar -> Direct Publish or Save Draft.
- **Flow 3 (Peer Battle Gameplay):** Launch 5-round battle -> 15s timer countdown -> select choice -> receive instant feedback -> view victory summary & DP reward.
- **Flow 4 (Daily Lightning Arena):** Enter Daily Lightning sprint -> answer 10 rapid-fire questions -> receive speed multiplier bonus DP.
- **Flow 5 (Deck Practice Session):** Open Decks page -> click Practice -> step through questions with full concept explanations -> increase deck mastery %.
- **Flow 6 (Social & Invites):** Send friend request -> accept incoming request -> challenge friend to duel -> generate +50 DP invite link.
- **Flow 7 (Leaderboard & Search):** Search global bar -> filter Monthly/All Time leaderboard -> trigger challenge against ranked scholar.